

AI-INTEGRATED CYBER SECURITY & CYBER FORENSICS PROGRAM

PROFESSIONAL TRAINING FRAMEWORK

A comprehensive, 60-hour case-driven curriculum bridging advanced network security, incident response execution, digital forensics, and responsible AI assisted automation workflows.



Duration: 60 Hours (27.5 Hours Concepts, 32.5 Hours Practical Labs)



Level: Beginner-to-Intermediate with Advanced Extension Paths



Delivery Format: Instructor-led, Lab-heavy, Case-driven



Executive Summary

This **60-hour program** is designed as a highly practical bridge between modern cybersecurity, advanced digital forensics, and operational AI workflows. Rather than positioning itself as a theory-only awareness program, learners actively work through realistic enterprise attack patterns, advanced defenses, and structured evidence investigations.

The curriculum caters concurrently to two distinct profiles:

Technical Participants:

Build deep execution capabilities in security operations, live incident triage, cloud audit investigation, cross-OS forensics, memory analysis, and AI automation.

Functional/Non-Technical Participants:

Learn how incidents surface, what digital evidence means under scrutiny, how risk decisions map to business governance, and how AI optimizes compliance and reporting defensibility.

Program Attribute	Summary Description
Total Duration	60 hours: 27.5 hours concepts, 32.5 hours labs, guided mini-projects, and capstone orientation.
Target Audience	Experienced IT/security professionals, SOC/IR teams, cloud/security engineers, GRC professionals, compliance auditors, technical managers, students, and career transitioners.
Core Outcome	Ability to systematically detect, investigate, isolate, document, and communicate high-impact security incidents using forensic discipline and responsible AI assistance.
Key Differentiator	Live, case-driven, tool-based training led by seasoned practitioners with published domain literature, enterprise consulting experience, and structured lab blueprints.
Portfolio Value	Hands-on mini-projects and comprehensive capstones yield ready-to-demonstrate career artifacts: technical forensics reports, evidence timelines, IOC lists, and AI validation logs.



Addressing Modern Real-World Threats

The program replaces outdated textbook exercises with contemporary threat vectors active across production enterprise architectures. Every lab module targets distinct attacker behaviors, defense controls, and evidence tracking.

Threat Area	Current Real-World Example	Course Response & Controls	AI / Forensics Pivot
Cloud & SaaS Compromise	Stolen session states, weak MFA parameters, over-permissive IAM policies, and exposed storage buckets.	Cloud audit log parsing, IAM hardening, least- privilege scoping, and storage evidence acquisition.	AI-assisted log summarization, anomaly clustering, and threat narrative drafting.
Zero Trust Gaps	Implicit trust inside flat legacy VPNs, unmanaged endpoint devices, and unmonitored account sharing.	Zero trust architecture building blocks, identity-centric controls, network segmentation, and policy enforcement points.	AI risk scoring across identity context, device posture, location, and asset sensitivity.
Credential Attacks	Automated password spraying, credential stuffing, and infostealer-driven corporate account takeovers.	Multi-factor authentication rules, conditional access, breach monitoring, and suspicious login triage.	AI identity anomaly detection, leaked-credential risk priority, and UEBA analysis.
Ransomware & Extortion	Spear-phishing entry vectors, edge device exploitation, lateral network movement, and multi-tier encryption.	Containment playbooks, backup integrity validation, endpoint evidence collection, and master incident reporting.	AI IOC extraction, automated timeline reconstruction, and asset grouping.
Supply-Chain Abuse	Third-party vendor compromises, malicious software updates, and untrusted open-source dependencies.	Vendor risk scoring, Software Bill of Materials (SBOM) awareness, tracking, and evidence boundaries.	AI-assisted dependency risk modeling and automated vendor incident ingestion tracking.
AI-Assisted Social Engineering	Highly convincing targeted phishing lures, professional voice cloning, and synthetic deepfake validation bypass.	Phishing header inspection, multi-channel verification workflows, and fraud control implementation.	NLP-based email risk scoring, deepfake indicator verification, and human-in-the-loop checks.
Vulnerability Exploitation	Active exploitation of known edge network equipment, application bugs, and exposed web infrastructure.	Patch prioritization structures, CISA KEV-style threat triage, and compensating control execution.	AI threat intelligence mapping matched against internal asset criticalities.
Insider Misuse & Leakage	Unauthorized data exfiltration, trace deletion, removal via physical storage, and unsanctioned cloud sharing.	Endpoint artifact analysis, host registry review, browser history forensics, and data visibility audits.	AI anomaly profiling, multi-source entity correlation, and event timeline generation.

Value Proposition & ROI Paths

The curriculum maps directly out to industry deliverables. Rather than isolated capture-the-flag exercises, learners employ a layered enterprise tool stack to build verifiable portfolios.

Layered Educational Tool Stack

The program replaces outdated textbook exercises with contemporary threat vectors active across production enterprise architectures. Every lab module targets distinct attacker behaviors, defense controls, and evidence tracking.

Category	Representative Tool Set	Strategic Skill Acquired
Forensic Case Platforms	Autopsy, The Sleuth Kit, OSForensics, DFIR-IRIS	Builds rigorous, legally defensible case discipline: data ingestion, structured tagging, and timeline building.
Acquisition & Imaging	FTK Imager, dd, Hardware Write Blockers, WinPmem	Teaches safe preservation methodologies, cryptographic hashing, and chain-of-custody execution.
Timeline & Log Analysis	Plaso / log2timeline, psort, Event Viewer, auditd	Reconstructs exact attacker activity sequences spanning complex multi-host enterprise networks.
Network Forensics	Wireshark, tcpdump, Zeek, NetFlow logs	Examines command-and-control communication, scanning phases, and active data exfiltration channels.
Memory & Malware Triage	Volatility, REMnux, PESTudio, oletools, Ghidra, YARA	Deconstructs memory captures, volatile processes, rootkits, document macros, and extracts operational IOCs.
Cloud & Identity Evidence	AWS CloudTrail, Azure Activity Logs, IAM Records	Links modern cloud configurations to concrete log artifacts for effective threat hunt visibility.
AI-Assisted Investigation	LLMs, Jupyter Notebooks, ML/NLP frameworks	Accelerates documentation, code translation, and analysis scale without compromising evidence integrity.

Targeted Professional Career ROI Paths

Current	Target Career TransitionPath	Core Portfolio ArtifactProduced
IT Support / SysAdmin	SOC Analyst, Incident Responder, Host Forensics Specialist	Endpoint Analysis Report & Ransomware Containment Playbook
Network Engineer	Network Security Engineer, Threat Hunter, Cloud Incident Handler	Full PCAP/DNS Investigation Brief & Network IOC Enrichment Set
Cloud / DevOps Engineer	Cloud Security Architect, DevSecOps Specialist, Zero Trust Engineer	Cloud Audit Log Analysis & Automated IAM Hardening Playbook
Developer / QA Engineer	Application Security Engineer, Security Automation Developer	Malware Triage Scripts & AI-Assisted Vulnerability Models
GRC / Audit Specialist	Cyber Risk Manager, Forensic Auditor, Incident Governance Lead	Defensible Evidence Handling Audit & Control Mapping Matrix
Team Lead / Technology Mgr	Security Director, Security Consultant, IR Coordinator	Executive Incident Response Briefing & Security Control Matrix

Cohort-Based Peer Architecture:

The program limits group sizing to small tracks of 3-5 peers to coordinate lab simulations, collaborate on case huddles, handle live mentor clinics, and showcase deliverables during the concluding Capstone Presentation Day.

Trainer Credibility & Positioning

The program is built entirely around field execution vectors. Instructional blueprints stem from 25+ years of cross-domain engineering, infrastructure hardening, enterprise databases, and active digital forensics consulting engagements

Verifiable Track Record

Core methodologies are backed by published works, including the Python for Cybersecurity Cookbook and the Digital Forensics Playbook (comprising over 100 practical analysis recipes), as well as foundational IEEE standard committee contributions.

Practical Orientation:

Training moves away from isolated game scenarios to mirror real-world multi-stage enterprise networks, multi-platform OS tracking, legal chain-of-custody workflows, and production-grade reporting rules.



60-Hour Structured Module Curriculum

The curriculum allocates specific balances of instructional theory alongside intense hand-on laboratory requirements. Every standalone module finishes with a distinct portfolio artifact.

Mod	Module Title & Core Objective	Theory	Lab	Total	Core Artifact Output
1	Program Orientation, Threat Landscape & AI Convergence: Establishing baseline paradigms, analyzing modern threat maps, and parsing responsible AI boundaries.	3.0h	2.0h	5.0h	Threat Architecture Map
2	Core Technical Foundations: Reviewing network structures, web application data paths, host file systems, and initializing safe virtual sandboxes.	2.5h	2.5h	5.0h	Evidence Source Sheet
3	Cyber Defense Operations: Structuring SOC alarm prioritization, parsing enterprise SIEM telemetry, mapping tactics to MITRE ATT&CK.	2.0h	3.0h	5.0h	SOC Triage Decisions & IOCs
4	AI for Cybersecurity: Engineering prompts, developing automated triage filters, evaluating model vulnerabilities and prompt injections.	2.0h	3.0h	5.0h	AI Triage Notebook
5	Forensic Readiness & Defensibility: Executing hashing verification, structuring acquisition plans, and tracking legal chain of custody.	2.5h	2.5h	5.0h	Defensible CoC Set
6	Endpoint & OS Forensics: Extracting artifacts from Windows registries, event records, Linux auditd tracking, and parsing MACB timestamps.	2.0h	3.0h	5.0h	Endpoint Artifact Record
7	Identity & Insider Threat Investigation: Uncovering brute-force trails, profiling credential abuse, and analyzing file deletion sequences.	2.0h	3.0h	5.0h	Insider Triage Case Log
8	Network, Web, Cloud & Zero Trust Analysis: Reviewing AWS/Azure control logging, hunting exfiltration in flow records, mapping vendor access vectors.	2.0h	3.0h	5.0h	Cloud Network Incident Brief
9	Malware, Ransomware & Memory Forensics: Extracting operational markers via Volatility, unpacking macro payloads, checking reputation records.	2.0h	3.0h	5.0h	Malware Analysis Ledger
10	Mobile, Email, Browser & Metadata Evidence: Reconstructing multi-source footprints via mobile packages, EXIF locations, and browser cache streams.	2.0h	3.0h	5.0h	Correlated Timeline Map
11	AI-Assisted DFIR Reporting & Dashboards: Building structured metrics dashboards, drafting defensible executive briefs, managing hallucination logs.	2.0h	3.0h	5.0h	AI-Validated Forensic Report
12	Capstone Studio & Project Defense: Conducting mock security briefings, finalizing student portfolios, peer reviews, and showcase presentation.	2.5h	2.5h	5.0h	Ready Interview Portfolio

Practical Project & Capstone Matrix

The culmination of the program relies on an intensive, 8-to-12-hour independent capstone track mimicking complex enterprise architecture compromises. Learners pick from one of twelve tailored options:

No	Capstone Blueprint Options	Strategic Techniques & Lab Tools Involved	Expected Capstone Deliverable
1	AI-Assisted Ransomware Case	Host event logs, ransom payloads, backup structures, Autopsy case suite, Plasotimelines, AI summary generation.	Complete multi-system forensic timeline and isolation brief.
2	Cloud Infrastructure Takeover	AWS CloudTrail, IAM permission nodes, Azure logs, VPC flow streams, impossible travel metrics, model abstraction.	Cloud threat vector report and Zero Trust remediation map.
3	Phishing-to-Malware Chain	Email headers, macro extraction, PE Studio triage, VirusTotal matching, YARA automation rules.	Full execution threat-chain audit and enterprise control advice.
4	Insider Exfiltration Audit	USB volume registries, MFT metadata tracking, web cache databases, user behavior correlation.	Defensible internal insider brief with complete chain-of-custody tracking.
5	Enterprise Zero Trust Assessment	Network maps, directory infrastructure, conditional policies, CISA Zero Trust model levels, AI scoring tables.	Prioritized security framework design and phased execution roadmap.
6	SOC Alert Noise Minimization	SIEM raw telemetry streams, asset data, deduplication rules, LLM pipeline validation mechanisms.	Operational analyst dashboard metrics and automated triage playbook.
7	Mobile & API Log Forensics	APK file manifests, MobSF security evaluations, web API payload logs, device artifacts, risk scoring.	Mobile security forensic report and developer validation guidelines.
8	Network Data Leak Hunting	Raw PCAP archives, DNS tunneling indicators, Zeek event formatting, connection behavior profiling.	Exfiltration vector brief and immediate containment plan.
9	Deepfake & Fraud Readiness	Synthetic multimedia scenarios, social engineering logs, operational control workflows.	Incident response validation playbook for executive fraud paths.
10	Forensic Readiness Design	Log storage lifetimes, compliance metrics, evidence asset lifecycles, automation control mapping.	Production-ready log lifecycle blueprint and infrastructure design.
11	AI System Prompt Exploitation	LLM query logs, data access controls, prompt injections, data leakage remediation.	AI app infrastructure risk report and model governance parameters.
12	Supply-Chain Breach Simulation	Vendor IAM credentials, software package updates, dependency trees, endpoint artifacts.	Supply-chain incident timeline and vendor audit framework checklist.

Open-Source & Free Tool Stack Reference

The curriculum builds real-world execution capacity leveraging widely-deployed open-source and free utilities, preparing participants for immediate workplace deployment without licensing dependencies.

Module Group	Core Integrated Tools	Recommended Extension Tool Set
Mod 1 &2	Ollama, Linux Core Utilities, Wireshark, tcpdump, Nmap, Sysinternals Suite	Open WebUI, LocalAI, SecurityOnion Framework, Zeek, Suricata, Arkime Packet Analyzer
Mod 3 &4	Wazuh SIEM, ELK Stack, KibanaUI, MISP Threat Intelligence, Sigma Detection Rules	TheHive Incident Platform, Cortex Analyser, ShuffleSOAR Automation, OpenSearch
Mod 5 & 6	Burp Suite Community, Postman API client, OWASP Juice Shop, DVWA Sandbox, Python Data Tools	crAPI, OWASP WebGoat Environment, JWT Toolkit, Scikit-learn, LangChain Framework, Haystack
Mod 7 & 8	Autopsy Forensic Suite, The Sleuth Kit (TSK), Volatility Memory Analyzer, KAPE Engine	Velociraptor Endpoint Agent, Chainsaw Event Triage, Plaso Timeline Engine, Timesketch UI
Mod 9 & 10	REMnux Environment, PESTudio Static Analyzer, VirusTotal API, Any.Run Sandbox, YARA Rule Engine	CAPA Code Capability Analyzer, FLOSS String Unpacker, YARA-X, Ghidra Disassembler, DFIR-IRIS

Assessment Architecture & Portfolio Metrics

Graduation from the program requires demonstrating practical mastery across five core metrics, verified by objective laboratory metrics, documented code artifacts, and oral project defenses.

Assessment Component	Weighting	Concrete Performance Evidence Expected
Lab Execution & Practical Worksheets	20%	Completed technical log files, tool interface screenshots, and baseline source mapping sheets.
Mini-Project Iteration Deliverables	25%	Structured vulnerability priority matrices, threat maps, and validation logs.
Strategic Capstone Execution	35%	Master incident investigation report, system remediation playbooks, and timeline data files.
Professional Governance Communication	10%	Executive summary brief clarity, clear incident communication, and mitigation rationale.
Operational Ethics & AI Compliance	10%	Proper evidence chain handling, data minimization compliance, and AI validation logs.

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